



ForgeGIS™

GPU-Accelerated Geospatial Primitives, Native to MCP

As of May 2026, the only commercially-supported GPU-accelerated geospatial library we have been able to identify that exposes its operation catalog to AI agents through a native Model Context Protocol server. 239 operations · 237 MCP-accessible · 68 MCP tools across two complementary surfaces. Operations run GPU-resident where the algorithm parallelizes on the device, with CPU fallbacks for the four sequential-by-nature ops (sink fill, flow accumulation, cost distance, nibble). Every operation is numerically deterministic and reachable from any MCP-capable agent — and fast enough that a multi-step pipeline returns in milliseconds, not a timeout.

237 / 239

operations MCP-accessible

two Java-only ops by design

68

MCP tools across two surfaces

60 intent + 5 primitive + 3 escape

8 tools

default agent surface (MINIMAL profile)

opt into the full 68 with one flag — keeps agents inside context budget

Two surfaces, one core

The MCP server exposes the catalog through two complementary access patterns. A pipeline DSL lets agents compose operations into multi-step workflows the server executes GPU-resident, with intermediate results staying on the device. New in 1.1, the pipeline engine runs non-linear DAGs — branching graphs across raster and vector stages. 60 curated intent tools wrap common analytic tasks as typed single-call endpoints with strict input schemas. Same operations underneath.

Deterministic by construction

The MCP layer is a parameter-marshaling shim over the same compute implementations the direct Java API exposes — not a parallel implementation. Numerical outputs depend on the operation and its parameters, not the invocation surface. The two operations that involve randomness (KMeansClassify, RandomForest) use a fixed deterministic default seed under MCP invocation, so agent runs are reproducible without per-call seed configuration.

Built to be embedded

Pure-Java Maven distribution with no first-party native code. The shaded MCP server jar drops into any host with a JDK and a GPU driver — no separate native install step, no CUDA-only runtime, no system-level GDAL, no platform-specific binaries to coordinate. Communicates over JSON-RPC on stdio per the MCP specification. Agents that spawn the server per session see roughly 800 ms median to first response on warm restart.

WHO THIS IS FOR

LLM agent platforms grounding spatial reasoning in real geometry

Agents that need to reason about where things are, what overlaps what, how far apart features sit, what a flood envelope looks like, or what's visible from a vantage point — and need the answer in milliseconds, not seconds. ForgeGIS exposes 237 operations through MCP without requiring the agent itself to understand projections, predicates, or pipeline composition.

Workflow and orchestration platforms

LangChain, AutoGen, MCP-native runners, and the broader class of platforms that compose tool calls into agent workflows. The dual-surface API — DSL for novel composition, intent tools for typed call shapes — lets the same library serve both "let the agent figure out the chain" and "give the agent a stable typed endpoint" workflows from one server install.

AI-native geospatial product builders

Founders and teams building product categories that didn't exist when the GIS stack was designed for human analysts at desktops — conversational geospatial intelligence, agent-driven site selection, AI-mediated remote sensing, autonomous mission planning. ForgeGIS is the compute substrate built for that thesis from the first commit.

PERFORMANCE HIGHLIGHTS

~800 ms

warm-restart cold-start to first tools/list response
(~3 s on first deploy, no OS cache)

9–65×

multi-stage GPU-resident pipelines vs. GDAL CLI,
measured warm at 1024²–4096² rasters

7.5×

geomean across 225 head-to-head records vs.
GDAL CLI, no category excluded (June 2026 run)

New in 1.1 — profile-gated discovery for agent context budgets

The server advertises a MINIMAL 8-tool surface by default (the 5 primitives plus 3 pipeline-escape tools), so an agent's tools/list stays small and stays inside context and rate limits. Opt into the full 68-tool STANDARD surface with `--profile standard` or an environment variable. Hidden tools remain callable by name — the profile governs discovery, not capability — so a curated agent loses nothing it can address directly.

Raw DE-9IM matrix for agent-encoded topology

Beyond the 8 OGC SFS named predicates (Intersects, Disjoint, Within, Contains, Equals, Touches, Crosses, Overlaps) and the 3 JTS extensions (Covers, CoveredBy, ContainsProperly), ForgeGIS exposes the raw 9-character DE-9IM intersection matrix per geometry-type pair through six matrix-computing operations. Agents reasoning about novel spatial relationships can encode topological queries that don't have a named predicate — useful when an agent needs a relationship the OGC vocabulary has no word for.

Validated against real-world data. ForgeGIS operations are exercised end-to-end against real digital elevation datasets — producing real elevation, viewshed, slope, aspect, contour, and intervisibility products through the same MCP surface an agent calls. Outputs are verified deterministic across repeated runs, so an agent-invoked result matches the scripted one byte for byte.

Build with ForgeGIS

A Technical Brief with the complete 239-operation catalog, MCP surface breakdown, validation methodology, and architectural deep-dive is available on request. Docker image and Maven coordinates available on request for evaluation.

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